

Spectral Embedding of Platonic Solids

In this page, we have not embedded for λ_{max} which has multiplicity 1 for every solid. Such plots give us a single point. We are unable to add them because the file fails to render too many interactive plots.

Tetrahedron

Eigenvalues:

`[-1. -1. -1. 3.]`

`= -1`

Eigenvectors for `= -1.0:`

Eigenvector 1:

`[ 0. -0.5774 0.7887 -0.2113]`

Eigenvector 2:

`[ 0. -0.5774 -0.2113 0.7887]`

Eigenvector 3:

`[-0.866 0.2887 0.2887 0.2887]`

Co-ordinates:

```
[[ 0.          0.          -0.8660254 ]
 [-0.57735027 -0.57735027  0.28867513]
 [ 0.78867513 -0.21132487  0.28867513]
 [-0.21132487  0.78867513  0.28867513]]
```

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We recover the original Tetrahedron (3-simplex). It shows that the eigenspace treats all the 4 vertices as separate entities.

Cube

Eigenvalues:

```
[-3. -1. -1. -1.  1.  1.  1.  3.]
```

= -3

Eigenvectors for = -3.0:

Eigenvector 1:

```
[ 0.3536 -0.3536  0.3536 -0.3536 -0.3536  0.3536 -0.3536  0.3536]
```

Co-ordinates:

```
[[ 0.35355339  0.          0.          ]
 [-0.35355339  0.          0.          ]
 [ 0.35355339  0.          0.          ]
 [-0.35355339  0.          0.          ]
 [-0.35355339  0.          0.          ]
 [ 0.35355339  0.          0.          ]
 [-0.35355339  0.          0.          ]
 [ 0.35355339  0.          0.          ]]
```

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Observation is that the 4 body diagonals collapse into one. Thus we get a line with two antipodal vertices (1-simplex). More accurately, if we consider a cube, the opposite vertices on the top face and their alternate vertices on bottom face ($\{2,4,5,7\}$ and $\{1,3,6,8\}$) collapse into one. In short, the collapsing vertices are those vertices which form a Tetrahedron within the Cube.

= -1

Eigenvectors for = -1.0:

Eigenvector 1:

```
[-0.5914  0.0852  0.3391  0.1672  0.3391  0.1672 -0.5914  0.0852]
```

Eigenvector 2:

```
[-0.1587  0.4701 -0.4761  0.1647 -0.4761  0.1647 -0.1587  0.4701]
```

Eigenvector 3:

```
[ 0.          0.3831  0.1826 -0.5656  0.1826 -0.5656  0.          0.3831]
```

Co-ordinates:

```
[[ -0.59144983 -0.15870444  0.          ]
 [ 0.08519785  0.47011757  0.38305456]
 [ 0.33909951 -0.47610733  0.18257419]
```

```
[ 0.16715247  0.16469419 -0.56562874]
[ 0.33909951 -0.47610733  0.18257419]
[ 0.16715247  0.16469419 -0.56562874]
[-0.59144983 -0.15870444  0.          ]
[ 0.08519785  0.47011757  0.38305456]]
```

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In this case, the antipodal pairs themselves collapse i.e the 4 body diagonals collapse into 4 points which form a Tetrahedron within the cube. Selecting exactly one vertex from each of the 4 body diagonals forms a Tetrahedron. The antipodal pairs are treated the same by symmetry.

$= 1$

Eigenvectors for $= 1.0000000000000002$:

Eigenvector 1:

```
[-0.0031  0.4864  0.5127  0.0232 -0.5127 -0.0232  0.0031 -0.4864]
```

Eigenvector 2:

```
[-0.6124 -0.2064  0.2013 -0.2047 -0.2013  0.2047  0.6124  0.2064]
```

Eigenvector 3:

```
[ 5.000e-04  3.096e-01 -2.675e-01 -5.767e-01  2.675e-01  5.767e-
01
-5.000e-04 -3.096e-01]
```

Co-ordinates:

```
[[ -3.13401981e-03 -6.12364215e-01  4.95824266e-04]
[  4.86358812e-01 -2.06365321e-01  3.09626324e-01]
[  5.12742375e-01  2.01286094e-01 -2.67542830e-01]
[  2.32495428e-02 -2.04712800e-01 -5.76673329e-01]
[ -5.12742375e-01 -2.01286094e-01  2.67542830e-01]
[ -2.32495428e-02  2.04712800e-01  5.76673329e-01]
[  3.13401981e-03  6.12364215e-01 -4.95824266e-04]
[ -4.86358812e-01  2.06365321e-01 -3.09626324e-01]]
```

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The original Cube is recovered. All vertices are separate.

Octahedron

Eigenvalues:

```
[-2. -2. -0. -0.  0.  4.]
```

$$= -2$$

Eigenvectors for $\lambda = -2.0000000000000004$:

Eigenvector 1:

[4.000e-04 -5.002e-01 4.998e-01 -5.002e-01 4.998e-01 4.000e-04]

Eigenvector 2:

[0.5774 -0.2883 -0.2891 -0.2883 -0.2891 0.5774]

Co-ordinates:

```
[[ 4.44410839e-04 5.77350098e-01 0.00000000e+00]
 [-5.00222057e-01 -2.88290178e-01 0.00000000e+00]
 [ 4.99777646e-01 -2.89059920e-01 0.00000000e+00]
 [-5.00222057e-01 -2.88290178e-01 0.00000000e+00]
 [ 4.99777646e-01 -2.89059920e-01 0.00000000e+00]
 [ 4.44410839e-04 5.77350098e-01 0.00000000e+00]]
```

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The antipodal vertices collapse to form a triangle (2-simplex). We see that the antipodal vertex pairs are treated the same by symmetry.

$$= 0$$

Eigenvectors for $\lambda = -2.993751573463213e-17$:

Eigenvector 1:

[-0.6972 0.1163 -0.0195 -0.1163 0.0195 0.6972]

Eigenvector 2:

[-0.0365 -0.1014 0.6988 0.1014 -0.6988 0.0365]

Eigenvector 3:

[-0.1121 -0.6901 -0.106 0.6901 0.106 0.1121]

Co-ordinates:

```
[[ -0.69720813 -0.03648073 -0.1121159 ]
 [ 0.1162742 -0.10141064 -0.6900697 ]
 [-0.01952251 0.6988455 -0.10598977]
 [-0.1162742 0.10141064 0.6900697 ]
 [ 0.01952251 -0.6988455 0.10598977]
 [ 0.69720813 0.03648073 0.1121159 ]]
```

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The 6 vertices are treated separately thus forming an Octahedron.

Dodecahedron

Eigenvalues:

```
[-2.2361 -2.2361 -2.2361 -2.      -2.      -2.      -2.      -0.      0.
 0.      0.      1.      1.      1.      1.      1.      2.2361 2.2361
 2.2361 3.      ]
```

$$= -\sqrt{5}$$

Eigenvectors for $\lambda = -2.2360679774997894$:

Eigenvector 1:

```
[-0.3873  0.2887 -0.1291 -0.1291  0.2887  0.2887 -0.1291 -0.1291  0.1291
 0.1291 -0.2887  0.1291  0.1291 -0.1291 -0.1291  0.1291  0.1291 -
0.2887
 0.3873 -0.2887]
```

Eigenvector 2:

```
[-0.      0.2528 -0.2373  0.3029 -0.081  -0.1718  0.359  -0.328  0.1217
-0.0251  0.1718 -0.359  0.328  -0.1217  0.0251  0.2373 -0.3029  0.081
0.      -0.2528]
```

Eigenvector 3:

```
[ 0.      -0.0524  0.2775 -0.2039  0.2451 -0.1928  0.0668 -0.1604  0.3443
-0.3643  0.1928 -0.0668  0.1604 -0.3443  0.3643 -0.2775  0.2039 -
0.2451
-0.      0.0524]
```

Co-ordinates:

```
[[ -3.87298335e-01 -6.55145067e-18  2.32990222e-36]
 [ 2.88675135e-01  2.52829231e-01 -5.23836466e-02]
 [-1.29099445e-01 -2.37306105e-01  2.77523235e-01]
 [-1.29099445e-01  3.02920297e-01 -2.03893666e-01]
 [ 2.88675135e-01 -8.10490470e-02  2.45148360e-01]
 [ 2.88675135e-01 -1.71780184e-01 -1.92764714e-01]
 [-1.29099445e-01  3.58995224e-01  6.67514984e-02]
 [-1.29099445e-01 -3.28037243e-01 -1.60389840e-01]
 [ 1.29099445e-01  1.21689119e-01  3.44274733e-01]
 [ 1.29099445e-01 -2.51169456e-02 -3.64283505e-01]
 [-2.88675135e-01  1.71780184e-01  1.92764714e-01]
 [ 1.29099445e-01 -3.58995224e-01 -6.67514984e-02]
 [ 1.29099445e-01  3.28037243e-01  1.60389840e-01]
 [-1.29099445e-01 -1.21689119e-01 -3.44274733e-01]
 [-1.29099445e-01  2.51169456e-02  3.64283505e-01]
 [ 1.29099445e-01  2.37306105e-01 -2.77523235e-01]
 [ 1.29099445e-01 -3.02920297e-01  2.03893666e-01]
```

```

[-2.88675135e-01  8.10490470e-02 -2.45148360e-01]
[ 3.87298335e-01  5.55111512e-17 -2.77555756e-17]
[-2.88675135e-01 -2.52829231e-01  5.23836466e-02]]

```

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A very symmetric distribution of vertices which lie on a sphere of radius 0.3873 units. More information can be found out using the Gram matrix. But an important point is that only five distinct angles occur between pairs of vertices.

= -2

Eigenvectors for $\lambda = -2.0000000000000004$:

Eigenvector 1:

```

[-0.4472  0.2981 -0.0745 -0.0745  0.2981  0.2981 -0.0745 -0.0745 -
0.0745
-0.0745  0.2981 -0.0745 -0.0745 -0.0745 -0.0745 -0.0745 -0.0745  0.2981
-0.4472  0.2981]

```

Eigenvector 2:

```

[ 0.          0.3316 -0.3318  0.1363 -0.136  -0.1956  0.1953 -0.3313  0.1357
 0.1958 -0.1956  0.1953 -0.3313  0.1357  0.1958 -0.3318  0.1363 -
0.136
-0.          0.3316]

```

Eigenvector 3:

```

[-0.          0.0338 -0.0855  0.3511 -0.2994  0.2656 -0.3173  0.018  0.2476
-0.2138  0.2656 -0.3173  0.018  0.2476 -0.2138 -0.0855  0.3511 -
0.2994
 0.          0.0338]

```

Eigenvector 4:

```

[-0.          0.0065  0.2775 -0.2293 -0.0547  0.0482  0.2358 -0.2905  0.3387
-0.3322  0.0482  0.2358 -0.2905  0.3387 -0.3322  0.2775 -0.2293 -
0.0547
 0.          0.0065]

```

Applying PCA (dim > 3 → reducing to 3D)

Co-ordinates:

```

[[-4.45937567e-18 -1.02620246e-18  1.26300820e-17]
[-2.97898190e-01  7.76657484e-02 -1.27811622e-01]
[ 1.83059541e-01 -3.10541013e-01  2.53964423e-01]
[ 2.57887955e-02  1.14932924e-01 -4.24935060e-01]
[ 8.90498538e-02  1.17942341e-01  2.98782260e-01]
[ 2.08848337e-01 -1.95608089e-01 -1.70970638e-01]

```

```

[-3.23686986e-01 -3.72671756e-02  2.97123438e-01]
[ 4.12736840e-01  1.55209516e-01  1.65882221e-03]
[-2.03888503e-01 -3.50817606e-01 -1.72629460e-01]
[-9.40096872e-02  4.28483354e-01  4.48178372e-02]
[ 2.08848337e-01 -1.95608089e-01 -1.70970638e-01]
[-3.23686986e-01 -3.72671756e-02  2.97123438e-01]
[ 4.12736840e-01  1.55209516e-01  1.65882221e-03]
[-2.03888503e-01 -3.50817606e-01 -1.72629460e-01]
[-9.40096872e-02  4.28483354e-01  4.48178372e-02]
[ 1.83059541e-01 -3.10541013e-01  2.53964423e-01]
[ 2.57887955e-02  1.14932924e-01 -4.24935060e-01]
[ 8.90498538e-02  1.17942341e-01  2.98782260e-01]
[ 1.08250435e-16 -2.36079120e-16 -1.86692695e-16]
[-2.97898190e-01  7.76657484e-02 -1.27811622e-01]]

```

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Antipodal vertices collapse and a symmetric shape is emerged. Due to the prominent triangle-looking structure within the projection, my intuition runs towards it being some n-simplex. But it is not. Since, to be a simplex vertex is to be adjacent to every other vertex.

$= 0$

Eigenvectors for $\lambda = -5.5161000268398e-16$:

Eigenvector 1:

```

[ 2.000e-04  3.383e-01  2.532e-01 -2.552e-01 -3.364e-01 -1.800e-
03
-8.320e-02 -2.534e-01 -2.551e-01 -8.300e-02  1.800e-03  8.320e-
02
 2.534e-01  2.551e-01  8.300e-02 -2.532e-01  2.552e-01  3.364e-
01
-2.000e-04 -3.383e-01]

```

Eigenvector 2:

```

[ 0.4285  0.0526 -0.2679 -0.2866  0.0734 -0.126 -0.1945 -0.1606  0.1419
 0.234  0.126  0.1945  0.1606 -0.1419 -0.234  0.2679  0.2866 -
0.0734
-0.4285 -0.0526]

```

Eigenvector 3:

```

[ 0.1222 -0.2364  0.1773  0.1268 -0.1898  0.4263 -0.0126 -0.2995  0.249
 0.1097 -0.4263  0.0126  0.2995 -0.249 -0.1097 -0.1773 -0.1268  0.1898
-0.1222  0.2364]

```

```

Eigenvector 4:
[-0.0384 -0.164    0.1808 -0.1914  0.213  -0.049   0.3938 -0.1424 -
0.2298
 0.3554  0.049  -0.3938  0.1424  0.2298 -0.3554 -0.1808  0.1914 -
0.213
 0.0384  0.164 ]

```

Applying PCA (dim > 3 → reducing to 3D)
Co-ordinates:

```

[[ 0.4291223  0.11513115  0.05047437]
 [ 0.0190251 -0.01595855 -0.18308664]
 [-0.25885554 -0.05588081  0.32488318]
 [-0.28255537  0.22860406 -0.06462994]
 [ 0.09326351 -0.27189585 -0.12764096]
 [-0.11228861  0.2878544   0.3107276 ]
 [-0.16559203 -0.32777666  0.19724221]
 [-0.17026676 -0.05925034 -0.37535754]
 [ 0.14656693  0.34373521 -0.01415557]
 [ 0.26353027 -0.21264551  0.24771658]
 [ 0.11228861 -0.2878544   0.3107276 ]
 [ 0.16559203  0.32777666 -0.19724221]
 [ 0.17026676  0.05925034  0.37535754]
 [-0.14656693 -0.34373521  0.01415557]
 [-0.26353027  0.21264551 -0.24771658]
 [ 0.25885554  0.05588081 -0.32488318]
 [ 0.28255537 -0.22860406  0.06462994]
 [-0.09326351  0.27189585  0.12764096]
 [-0.4291223  -0.11513115 -0.05047437]
 [-0.0190251  0.01595855  0.18308664]]

```

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Each vertex is distinct in this concave polyhedron. The distribution of vertices looks mildly chaotic but from a certain perspective, we see a symmetric pattern. The adjacent vertices are not equidistant. That could be due to the loss of information from 4D to 3D projection.

$= 1$

Eigenvectors for $= 1.0000000000000002$:

```

Eigenvector 1:
[ 0.0005 -0.4371 -0.0814  0.2351  0.2762  0.1614  0.0403 -0.3562  0.0406
 0.1206  0.1614  0.0403 -0.3562  0.0406  0.1206 -0.0814  0.2351  0.2762
 0.0005 -0.4371]

```

Eigenvector 2:

```
[-0.0171 -0.0122  0.166  -0.2511 -0.3065  0.3017 -0.1106 -0.1611 -  
0.0383  
 0.4293  0.3017 -0.1106 -0.1611 -0.0383  0.4293  0.166  -0.2511 -  
0.3065  
-0.0171 -0.0122]
```

Eigenvector 3:

```
[-0.4984 -0.1536  0.1923  0.1948 -0.1535 -0.1913  0.1561  0.1526  0.15  
 0.151  -0.1913  0.1561  0.1526  0.15  0.151  0.1923  0.1948 -  
0.1535  
-0.4984 -0.1536]
```

Eigenvector 4:

```
[-0.0361 -0.1846 -0.4183 -0.275  -0.0117  0.1601  0.1549  0.2699  0.2995  
 0.0413  0.1601  0.1549  0.2699  0.2995  0.0413 -0.4183 -0.275  -  
0.0117  
-0.0361 -0.1846]
```

Eigenvector 5:

```
[ 0.0043  0.0334  0.0621 -0.1345  0.2368 -0.2659 -0.4333 -0.0329  0.367  
 0.1631 -0.2659 -0.4333 -0.0329  0.367  0.1631  0.0621 -0.1345  0.2368  
 0.0043  0.0334]
```

Applying PCA (dim > 3 → reducing to 3D)

Co-ordinates:

```
[[ -0.01309071 -0.02197957 -0.49073227]  
[ -0.06321089  0.35946546 -0.13922547]  
[ -0.21002275  0.30172833  0.11389652]  
[ -0.11858759  0.1637438  0.11776621]  
[ -0.30700925 -0.20403105 -0.14368668]  
[  0.35712943 -0.17741397 -0.20782011]  
[  0.39844441  0.06604652  0.14755637]  
[  0.15990257  0.0797167  0.23761027]  
[ -0.17533095 -0.34579529  0.22927938]  
[ -0.02822427 -0.22148093  0.13535579]  
[  0.35712943 -0.17741397 -0.20782011]  
[  0.39844441  0.06604652  0.14755637]  
[  0.15990257  0.0797167  0.23761027]  
[ -0.17533095 -0.34579529  0.22927938]  
[ -0.02822427 -0.22148093  0.13535579]  
[ -0.21002275  0.30172833  0.11389652]  
[ -0.11858759  0.1637438  0.11776621]  
[ -0.30700925 -0.20403105 -0.14368668]  
[ -0.01309071 -0.02197957 -0.49073227]  
[ -0.06321089  0.35946546 -0.13922547]]
```

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We see that the 20 vertices collapse into 10. The antipodal pairs are once again treated the same by eigenspace due to symmetry. The 3D projection of the 5D embedding is not a recognized structure as per my limited knowledge.

$$= \sqrt{5}$$

Eigenvectors for $\lambda = 2.2360679774997902$:

Eigenvector 1:

```
[ 0.0012  0.2585  0.3036  0.0741 -0.1128 -0.1431  0.0251  0.2732  0.3275
 0.3462  0.1431 -0.0251 -0.2732 -0.3275 -0.3462 -0.3036 -0.0741  0.1128
-0.0012 -0.2585]
```

Eigenvector 2:

```
[-0.3863 -0.2884 -0.1136 -0.1035 -0.2721 -0.3034 -0.1542 -0.145  0.1185
 0.1379  0.3034  0.1542  0.145  -0.1185 -0.1379  0.1136  0.1035  0.2721
 0.3863  0.2884]
```

Eigenvector 3:

```
[ 0.0271  0.0027  0.212  0.3658  0.2515 -0.1936 -0.3544 -0.2331 -
0.1695
 0.1056  0.1936  0.3544  0.2331  0.1695 -0.1056 -0.212  -0.3658 -
0.2515
-0.0271 -0.0027]
```

Co-ordinates:

```
[[ 0.00116566 -0.38634426  0.02714321]
 [ 0.25847264 -0.28841725  0.00271737]
 [ 0.30355606 -0.11360749  0.21200721]
 [ 0.07411217 -0.10349612  0.36578127]
 [-0.11277537 -0.27205672  0.25152904]
 [-0.14309076 -0.30341806 -0.19355235]
 [ 0.02506084 -0.15423984 -0.35437555]
 [ 0.27324067 -0.14496883 -0.23307418]
 [ 0.32745124  0.11849694 -0.16951154]
 [ 0.34618718  0.13787931  0.10556388]
 [ 0.14309076  0.30341806  0.19355235]
 [-0.02506084  0.15423984  0.35437555]
 [-0.27324067  0.14496883  0.23307418]
 [-0.32745124 -0.11849694  0.16951154]
 [-0.34618718 -0.13787931 -0.10556388]
 [-0.30355606  0.11360749 -0.21200721]
 [-0.07411217  0.10349612 -0.36578127]
 [ 0.11277537  0.27205672 -0.25152904]
```

```
[-0.00116566  0.38634426 -0.02714321]
[-0.25847264  0.28841725 -0.00271737]]
```

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We get a Dodecahedron.

Icosahedron

Eigenvalues:

```
[-2.2361 -2.2361 -2.2361 -1.      -1.      -1.      -1.      -1.      2.2361
 2.2361  2.2361  5.      ]
```

$$= -\sqrt{5}$$

Eigenvectors for $= -2.2360679774997907$:

Eigenvector 1:

```
[ 0.4999 -0.2331 -0.2148 -0.2248 -0.2169  0.2282  0.2248  0.2169 -
0.2282
 0.2331  0.2148 -0.4999]
```

Eigenvector 2:

```
[-2.000e-04  7.420e-02  1.994e-01  4.424e-01 -3.191e-01  3.964e-
01
-4.424e-01  3.191e-01 -3.964e-01 -7.420e-02 -1.994e-01  2.000e-
04]
```

Eigenvector 3:

```
[ 0.0109  0.4361 -0.4051  0.0609 -0.318  -0.2018 -0.0609  0.318  0.2018
-0.4361  0.4051 -0.0109]
```

Co-ordinates:

```
[[ 4.99881784e-01 -2.27616573e-04  1.08697094e-02]
[-2.33107981e-01  7.42027353e-02  4.36067224e-01]
[-2.14759687e-01  1.99384833e-01 -4.05122161e-01]
[-2.24783325e-01  4.42446368e-01  6.09398729e-02]
[-2.16889394e-01 -3.19079090e-01 -3.18036988e-01]
[ 2.28229263e-01  3.96445881e-01 -2.01846643e-01]
[ 2.24783325e-01 -4.42446368e-01 -6.09398729e-02]
[ 2.16889394e-01  3.19079090e-01  3.18036988e-01]
[-2.28229263e-01 -3.96445881e-01  2.01846643e-01]
[ 2.33107981e-01 -7.42027353e-02 -4.36067224e-01]
[ 2.14759687e-01 -1.99384833e-01  4.05122161e-01]
[-4.99881784e-01  2.27616573e-04 -1.08697094e-02]]
```

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Here, all the vertices are distinctly spread out at a distance of 0.5 units from the origin i.e they lie on a sphere of radius 0.5 units. The 6 antipodal pairs are clearly visible as they still remain the pairs of vertices farthest apart from each other. A beautiful concave and symmetric-looking polyhedron is recovered. Intuition points towards this shape being another form of Icosahedron itself. only three distinct angles occur between pairs of vertices

$= -1$

Eigenvectors for $\lambda = -1.0000000000000002$:

Eigenvector 1:

```
[ 2.000e-04 -2.919e-01 -4.691e-01  2.831e-01  2.238e-01  2.539e-
01
 2.831e-01  2.238e-01  2.539e-01 -2.919e-01 -4.691e-01  2.000e-
04]
```

Eigenvector 2:

```
[ 0.1028 -0.5464  0.3768 -0.0623 -0.0708  0.1999 -0.0623 -0.0708  0.1999
-0.5464  0.3768  0.1028]
```

Eigenvector 3:

```
[-0.6326  0.0294  0.1955  0.0871  0.1894  0.1312  0.0871  0.1894  0.1312
 0.0294  0.1955 -0.6326]
```

Eigenvector 4:

```
[ 0.0767 -0.1094  0.0201 -0.2927  0.5707 -0.2654 -0.2927  0.5707 -
0.2654
-0.1094  0.0201  0.0767]
```

Eigenvector 5:

```
[-0.0012  0.1417 -0.1263 -0.4893  0.0013  0.4739 -0.4893  0.0013  0.4739
 0.1417 -0.1263 -0.0012]
```

Applying PCA (dim > 3 → reducing to 3D)

Co-ordinates:

```
[[ 0.00581536  0.0292582  0.02353414]
[-0.43561786  0.11811885  0.19572898]
[ 0.3316647  -0.46011088  0.25493581]
[-0.08808949  0.15849584  0.26744772]
[-0.20277631 -0.26673727 -0.52829695]
[ 0.38900359  0.42097526 -0.2133497 ]
[-0.08808949  0.15849584  0.26744772]
[-0.20277631 -0.26673727 -0.52829695]
[ 0.38900359  0.42097526 -0.2133497 ]
```

```
[-0.43561786  0.11811885  0.19572898]
[ 0.3316647  -0.46011088  0.25493581]
[ 0.00581536  0.0292582   0.02353414]]
```

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The original 5D embedding is projected onto 3D system using PCA. The most fascinating part is that this is a projection of a 5-simplex.

$$= \sqrt{5}$$

Eigenvectors for $\lambda = 2.236067977499791$:

Eigenvector 1:

```
[ 0.0159 -0.2173  0.3055 -0.0387 -0.4299 -0.4158  0.0387  0.4299  0.4158
 0.2173 -0.3055 -0.0159]
```

Eigenvector 2:

```
[-0.0321  0.3712  0.3186 -0.4584 -0.109   0.1942  0.4584  0.109  -
0.1942
-0.3712 -0.3186  0.0321]
```

Eigenvector 3:

```
[-0.4987 -0.255  -0.235  -0.1959 -0.2308  0.1985  0.1959  0.2308 -
0.1985
 0.255   0.235   0.4987]
```

Co-ordinates:

```
[[ 0.0158513  -0.0321258  -0.49871502]
 [-0.2172541  0.37118156 -0.25499871]
 [ 0.30545229  0.31855361 -0.23499468]
 [-0.03865737 -0.45840558 -0.19588245]
 [-0.42992688 -0.10900563 -0.23082602]
 [-0.41583195  0.19415944  0.19845882]
 [ 0.03865737  0.45840558  0.19588245]
 [ 0.42992688  0.10900563  0.23082602]
 [ 0.41583195 -0.19415944 -0.19845882]
 [ 0.2172541  -0.37118156  0.25499871]
 [-0.30545229 -0.31855361  0.23499468]
 [-0.0158513  0.0321258   0.49871502]]
```

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Icosahedron is recovered.

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